AMITY UNIVERSITY

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Amity University Uttar Pradesh, Greater Noida

Minor Project (ETMN100)

WEEKLY PROGRESS REPORT (WPR) – 2

For the week commencing:	**27/07/2026 – 02/08/2026**
Enrollment Number:	**A41105223039**
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Program:	**B.Tech CSE**
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Project Title:	**WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar**

Abstract

Access to safe drinking water remains a critical public health concern across rural India, particularly in the states of Uttar Pradesh and Bihar, where chemical and microbiological contamination of groundwater sources affects millions of households. Despite the Jal Jeevan Mission (JJM) generating extensive water quality testing data at the village and habitation level, this information remains largely inaccessible to local district and block-level officials who are responsible for prioritising remediation efforts and allocating limited field resources. This project proposes WaterTriage, an open, data-driven triage system designed to bridge this accessibility gap. The system ingests publicly available JJM water quality datasets and processes them through a structured backend pipeline built on Python, FastAPI, SQLAlchemy, and PostgreSQL, fully containerised with Docker for reproducibility and ease of deployment. At its core lies a contamination severity scoring engine that evaluates individual water sources against Bureau of Indian Standards (BIS) 10500 permissible limits for parameters including arsenic, fluoride, nitrate, iron, turbidity, and E. coli, applying differential weights to reflect the relative health risk of each contaminant rather than treating all exceedances as equivalent. A recurrence detection module distinguishes persistent contamination patterns from isolated testing anomalies across the 2022–2025 data span, enabling more informed intervention decisions that account for temporal reliability. Ranked priority lists are exposed through RESTful API endpoints, filterable by state, district, block, contamination type, and severity band, with a lightweight dashboard providing geospatial visualisation for demonstration purposes. Data acquisition is currently being pursued through multiple parallel channels including the AIKosh open data portal, the National Water Data Portal (CKAN), and the WQMIS public reporting system. The expected outcome is an independently reproducible prototype that demonstrates how analytics built on government open data can support ground-level water safety planning without reliance on proprietary or internal government systems that remain inaccessible to local stakeholders.

Keywords: Water Quality; Jal Jeevan Mission; Contamination Severity Scoring; BIS 10500; FastAPI; Intervention Prioritization; Recurrence Detection